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The integral in this exercise converges. Evaluate the integral without using a table.

$$\int_1^{\infty} \frac{1}{x^{1.00004}} dx$$

Recall that, for Type I improper integrals, if $f(x)$ is continuous on $[a, \infty)$, then $\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$. So $\int_1^{\infty} \frac{1}{x^{1.00004}} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^{1.00004}} dx$.

Integrate the function.

$$\begin{aligned} \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^{1.00004}} dx &= \lim_{b \rightarrow \infty} \left[\frac{x^{-1.00004+1}}{-1.00004+1} \right]_1^b && \text{Use } \int x^n dx = \frac{1}{n+1} x^{n+1} + C. \\ &= \lim_{b \rightarrow \infty} \left[-\frac{25,000}{x^{0.00004}} \right]_1^b \end{aligned}$$

Substitute the limits of integration.

$$\lim_{b \rightarrow \infty} \left[-\frac{25,000}{x^{0.00004}} \right]_1^b = \lim_{b \rightarrow \infty} -\frac{25,000}{b^{0.00004}} - \left(-\frac{25,000}{1^{0.00004}} \right)$$

Evaluate the limit of the expression to find the value of the integral.

$$\begin{aligned} \lim_{b \rightarrow \infty} -\frac{25,000}{b^{0.00004}} - \left(-\frac{25,000}{1^{0.00004}} \right) &= 0 - (-25,000) \\ &= 25,000 \end{aligned}$$

Therefore, $\int_1^{\infty} \frac{1}{x^{1.00004}} dx = 25,000$.